HealthConnect Rwanda

Final Analytics Integration Readiness Record

Programme:	AnalystLab Africa Experience Lab	**Week:**	Week 8
Track:	Data Analytics	**Project:**	HealthConnect Clinic Experience Lab
Project Title:	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI	**Prepared by:**	Victoria Kosgei

1. FINAL COMPONENT

My final component is the **HealthConnect Rwanda Healthcare Analytics and Business Intelligence Dashboard**, developed using Microsoft Power BI. The component includes a six-page interactive executive dashboard covering:

- Executive Overview
- Clinical Outcomes
- Patient Demographics
- Financial Performance
- Operational Performance
- Patient Experience

It provides validated analytical outputs, KPI measures, business insights, and recommendations based on the available healthcare datasets.

2. BUSINESS PROBLEM ADDRESSED

HealthConnect Clinic seeks to reduce missed appointments, improve patient appointment attendance, make better use of available appointment slots, and provide effective administrative support. The analytics component supports these objectives by providing visibility into appointment patterns, patient demographics, clinical records, financial performance, and data quality. The current dataset does not contain attendance status, so the dashboard cannot directly measure attendance or no-show rates.

3. CURRENT COMPONENT STATUS

Requirement	Status
Six-page Power BI dashboard	**Developed**
KPI calculations	**Developed and reviewed**
Financial KPI validation	Completed for the reviewed measures
Patient demographic validation	Completed
Appointment visit-type validation	Completed
Dashboard functionality testing	Passed tested interactions
Data quality validation	Completed; limitations documented
Final presentation packaging	In progress
Final cross-track integration	Requires final output exchange and evidence

4. WEEK 7 IMPROVEMENTS

Week 7 strengthened the analytics component through:

- Validation of gross billed revenue and payments collected against source tables.
- Review of billing-to-appointment matching.
- Validation of the three appointment visit types.
- Correction and confirmation of patient gender percentages.
- Testing of slicers, KPI responsiveness, chart cross-filtering, navigation, and reset functionality.
- Documentation of financial, facility mapping, attendance, and resource utilization limitations.
- Review of Data Science testing findings relevant to the wider solution.

5. DEPENDENCIES

Tracks My Component Depends On

Track	Dependency
Data Science	Final model findings and interpretation requirements
ML Engineering	Confirmation of the model or workflow integration requirements
Generative AI	Confirmation of administrative assistant capabilities and limitations
Project Management	Final integration sequence, presentation requirements, and readiness coordination

Tracks That Depend on My Output

Track	Analytics Output
Data Science	Validated analytical findings and data limitations to support model interpretation
Project Management	Final KPIs, business insights, recommendations, and analytics readiness status
ML Engineering	Relevant data quality findings and analytical requirements
Generative AI	Approved analytical context and relevant healthcare findings, where required and supported

6. OUTPUT I WILL PROVIDE

- Final Power BI dashboard & validated KPI summary
- Key analytical findings, business insights, and recommendations
- Analytical limitations, dashboard screenshots, and supporting evidence
- Analytics presentation slides and final analytics documentation

7. OUTPUT REQUIRED FROM OTHER TRACKS

- **Data Science:** Final model evaluation findings and interpretation.
- **ML Engineering:** Final workflow or pipeline integration details.
- **Generative AI:** Assistant capabilities, approved knowledge sources, limitations, and demonstration requirements.
- **Project Management:** Final integration walkthrough sequence and submission requirements.

8. EVIDENCE OF READINESS

Week 6 Power BI dashboard; Week 7 KPI validation record; Data quality validation record; Dashboard functionality testing record; Analytical findings validation; Data limitations and refinement record; Actual dashboard screenshots; Cross-track testing documentation, where supported by evidence.

9. REMAINING LIMITATIONS

1. 55,000 billing records are unmatched to appointment records.
2. Approximately 5% facility mapping limits facility-level analysis.
3. The appointment dataset does not contain an attendance status field.
4. Bed occupancy history is unavailable; reliable room occupancy data is unavailable.
5. Equipment usage and downtime history are insufficient for utilization analysis.
6. Financial reconciliation and Revenue Summary differences require further investigation.
7. The Data Science emergency prediction model did not meet its reported release criteria.

10. READINESS ASSESSMENT

The analytics dashboard and its documented findings are prepared for final presentation and integration review. The component is ready to contribute to the wider HealthConnect solution as a decision-support output. Final end-to-end integration depends on the confirmation and exchange of outputs from the other HC-POD tracks.

> **Readiness Status:** Analytics component prepared for final integration review; remaining dependencies and data limitations documented.